Chapter 2 exercises

DM02-E01

Use a full truth table to determine whether $((p \to q) \land (q \to r)) \to (p \to r)$ is a tautology. Explain every critical row.

DM02-E02

Determine whether $\neg p, p \lor q, q \to r \therefore r$ is valid. Give inference rules if valid, or a counterexample valuation otherwise.

DM02-E03

Let the domain be the real numbers, with $x, y \in \mathbb{R}$, and let P be a binary predicate on the reals. Put $\neg \exists x \forall y (P(x, y) \land (x < y))$ into negation form with all quantifiers in front. Explain the original statement in words.

DM02-E04

Let n be an integer. Prove that if $3n + 2$ is odd, then n is odd. Name your proof method.

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DM02-E05

Find an integer n refuting "if n^2 has remainder one modulo four, then n is prime." Explain how it meets the definition of a counterexample.

DM02-E06

Over the real numbers, determine the truth of $\forall x \exists y (x + y = 0)$ and $\exists y \forall x (x + y = 0)$. Explain witnesses and their dependence.

DM02-E07

Simplify $\neg((p \lor q) \land \neg r) \lor q$ using only logical laws, naming the law at every step.

DM02-E08

Over a nonempty domain, prove $\forall x (P(x) \to Q(x)), \exists x P(x) \therefore \exists x Q(x)$. Mark universal specification, modus ponens, and existential generalization.

DM02-E09

Does $\forall x(P(x) \lor Q(x)) \to (\forall x P(x) \lor \forall x Q(x))$ hold over integers? If not, provide predicates in one common counterexample interpretation.

DM02-E10

Prove that $\sqrt{2}$ is irrational. State the contradiction assumption, reduced-fraction condition, each use of the parity lemma, and the final contradiction.

DM02-E11

Why cannot $\exists x P(x)$ and $\exists x Q(x)$ share one fresh name c to infer $\exists x(P(x) \land Q(x))$? Give a counterexample on one domain and state the conclusion restriction for existential elimination.

DM02-E12

Give the full truth table of $\varphi = (p \leftrightarrow q) \lor r$, then its canonical DNF and CNF. Express $p \lor q$ with NAND alone, and explain how a general DNF yields a NAND formula for any Boolean function of $n \geq 1$ variables.

References and attribution

Lin Cing, handwritten discrete mathematics notes. Ralph P. Grimaldi, Discrete and Combinatorial Mathematics, 5th ed., Ch. 2.

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